Natural Language Processing

Project Report

NewsBrief

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(group of 2 members)

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1. Abstract

This project tackles the problem of information overload, due to the growing amount of digital news. Readers are often unable to efficiently process lengthy news articles, creating a need for automated systems that can generate concise and meaningful summaries. To meet this challenge, this project is designed to create NewsBrief, an abstractive news summarization system that is based on a Bidirectional and Auto-Regressive Transformers (BART) model. The proposed system learns the semantic meaning and the contextual relations in the text to be used in generating human-like summaries as opposed to extractive methods that directly copy sentences from the source documents. The facebook/bart-large-cnn model was implemented and evaluated using the Inshorts News Summary dataset, with performance assessed using ROUGE evaluation metrics against human-written reference summaries. The system was also implemented as an interactive Streamlit web application that was integrated with NewsAPI to get the summary of the news as it happens. The experimental results show that transformer-based abstractive summarisation can successfully decrease reading time without sacrificing the main information and readability of news articles. This project illustrates the implementation of NLP techniques to make news consumption scalable and accessible.

2. Introduction

Millions of stories are published each day on digital platforms but most readers find themselves reading just a fraction of the content available, not because they don't have an interest, but because they don't have the time. This disconnect between the amount of information generated and the ability to process it has given rise to a well-documented phenomenon known as information overload — a state in which excessive information actively prevents meaningful engagement rather than enabling it. As readers grow weary of the decision making process, they miss out on important events and eventually turn off to the news altogether. Students, professionals and general readers are all impacted, and the demand for a more timely and effective means of information is as critical as ever.

The main issue this project attempts to solve is the lack of a reliable, automated system capable of summarising long news articles into concise summaries without human editorial intervention. Current solutions are expensive, slow and unable to scale across the thousands of articles produced every day, since they either demand readers to engage with complete articles or rely on human editors, as platforms like Inshorts do. As a result, there is a clear and unmet demand for an NLP-powered solution that can automatically replicate the quality of human summarisation.

This is quite a challenge to solve using automation. Summarizing news requires a system to understand context, discern the most newsworthy facts, discard irrelevant background detail and produce coherent, fluent output that retains the original meaning. Traditional extractive approaches like TextRank choose and copy sentences from the original, yielding disjointed and unnatural summaries. Abstractive summarization processes semantic meaning and rephrases it into new text, yielding more natural and readable output, but demands the language understanding capability that only modern transformer-based architectures can deliver.

Solving this problem has numerous ramifications — news organisations could expand editorial operations without corresponding cost increases, readers could stay informed in a fraction of the time, and news would be more accessible for those with cognitive impairments or language barriers. This project directly tackles this challenge by developing NewsBrief, investigating the feasibility of achieving high-quality abstractive summaries using the BART transformer model for automated news summarization at scale.

3. Background

3.1 Challenge 1: Generating Accurate, Human-Like Summaries

Reducing the length of an article is not a fundamental problem for automated news summarisation, but rather the task of producing a fluent, coherent and factually accurate summary. A summarization system is required to extract the most salient information, interpret the interactions among facts in the source text, eliminate irrelevant background information, and even generate completely new text that would be naturally understood — while not misrepresenting the source. This is particularly tricky in abstractive summarization, where the model generates new sentences rather than copying existing ones, introducing the risk of hallucination — where the model produces plausible-sounding but factually incorrect statements. This tension was first pointed out by Rush et al. (2015), who found that attention-based encoder-decoders generated more readable summaries than extractive models, but were less well-known to be unable to maintain factual consistency. Lewis et al. (2020) tackled this by introducing a denoising pre-training objective to BART to help build more factual knowledge. But BART is not immune from hallucination, especially when presented with domain-specific content that is not similar to the content it was trained on. This tension between fluency and factual accuracy represents the core technical difficulty that any abstractive summarization system must navigate.

3.2 Challenge 2: Domain Mismatch and Dataset Limitations

One of the challenges in training and evaluating summarization models is the domain mismatch. Large pre-trained models like BART are typically fine-tuned on Western news datasets such as CNN/DailyMail, which predominantly feature long-form English journalism from American and British outlets. See et al. (2017) showed that models trained on CNN/DailyMail struggled to generalize to articles from different domains and writing styles - directly relevant to our project, as the Inshorts dataset consists primarily of Indian news articles differing significantly in cultural context, named entities and linguistic patterns. Furthermore, Inshorts reference summaries are constrained to exactly 60 words — a far stricter length requirement than CNN/DailyMail's variable-length summaries. Such discrepancy between the training and evaluation datasets directly affect the ROUGE scores as the model produces text with different style even if the semantic content is right, as reported by Nallapati et al. (2016). This would need domain-specific fine-tuning on a large corpus of news collected from India, which would be an interesting research direction to take.

3.3 Challenge 3: Multi-Topic Articles and Focus Selection

A third important problem is the content of news articles which cover several topics, events or points of view in a single article. A human editor editing such an article has the option to make a conscious editorial decision as to what aspect to focus on. An automated model lacks this editorial sense and uses only statistical patterns to decide on importance. Liu and Lapata (2019) showed that even state-of-the-art transformer models struggle with multi-topic documents, as attention mechanisms tend to disproportionately weight content appearing early in the article. This was clearly observed in our results — in an article about Christopher Nolan's Dunkirk, BART focused on cinematic qualities while the Inshorts reference focused on its release and critical reception. The cases here show the basic challenge of machine summarization and human editorial judgement and point to the need for further advanced approaches to content selection in future work.

4. Methodology

4.1 Why Abstractive Summarization?

Text summarization has two main approaches: extractive and abstractive. Extractive methods such as TextRank (Mihalcea & Tarau, 2004) select and rank key sentences from the original text to form a summary. While this preserves factual accuracy, the resulting summaries are often fragmented and lack coherence because sentences are not rephrased or logically connected. Abstractive summarization instead generates new sentences by understanding the semantic meaning of the source and producing a condensed rephrasing. This is better suited for NewsBrief's goal of creating natural, Inshorts-style summaries. However, it is more challenging due to the need for deep language understanding and risks of generating inaccurate or inconsistent information.

4.2 Transformer Architecture

Our solution's theory is based on the transformer architecture introduced by Vaswani et al. (2017) in their seminal paper "Attention is All You Need. This transformer eliminates the need for recurrent neural networks, enabling the model to consider the relevance of each word the input sequence relative to other words as it is reading the entire sequence, whether close or far apart within the text. This can be useful for news summarisation, because the important information might be spread across a lengthy article, instead of being summarised at the start. The encoder processes the full input article and generates contextual representations of each token, while the decoder generates the summary sequentially, attending to relevant encoded information at each step. All modern sequence-to-sequence summarization models, including BART, are based on this encoder-decoder architecture.

4.3 BART - Bidirectional and Auto-Regressive Transformers

BART (Lewis et al., 2020) is a sequence-to-sequence Transformer model designed for generative tasks such as summarization. It consists of a bidirectional encoder, which captures contextual representations by attending to both left and right context (similar to BERT), and an autoregressive decoder, which generates output text sequentially one token at a time (similar to GPT). BART is given the denoising objective, which means the model learns to reconstruct the original text from different corrupted versions of the text, including by masking, deleting, permuting, and rotating the text tokens, and this gives BART a deep understanding of the structure and the semantics of the language (Lewis et al., 2020). We will leverage the facebook/bart-large-cnn variant that was fine-tuned on the CNN/DailyMail news summarization dataset, and will be directly usable on our news summarization task without the need to train it from scratch.

4.4 Beam Search Decoding

During inference, BART uses beam search, a heuristic decoding algorithm that maintains multiple candidate sequences at each generation step. Instead of greedily selecting the most probable next token, beam search explores a fixed number of hypotheses (beams) to improve overall sequence quality (Freitag & Al-Onaizan, 2017). In this project, a beam size of 4 is used, allowing the model to maintain four candidate outputs at each step. A length penalty of 2.0 is applied to discourage overly short summaries, and a no-repeat n-gram constraint of size 3 is used to reduce repetition and improve fluency. This configuration prioritizes summary quality over computational efficiency, which is appropriate for offline or API-based inference in the NewsBrief application.

4.5 Model Selection and Deployment Considerations

A key design consideration is balancing model performance with computational feasibility. The facebook/bart-large-cnn model contains approximately 406 million parameters, making it computationally expensive but highly effective for abstractive summarization. Rather than training a model from scratch, this project leverages a pre-trained and fine-tuned checkpoint, enabling strong performance under limited computational resources (Kaggle T4 GPU). This aligns with practical deployment constraints commonly encountered in real-world NLP systems, where efficiency and scalability are critical.For deployment, inference is handled via the HuggingFace Inference API, which offloads computation to external infrastructure. This removes the need for dedicated GPU hardware and enables scalable access to the summarization system.

4.6 ROUGE Evaluation Metric

Lin (2004) proposed a metric called ROUGE (Recall-Oriented Understudy for Gisting Evaluation) which is used to assess the quality of generated summaries. There are three variants of ROUGE metrics that measure the similarities and differences between a machine-generated summary and one or more human-written reference summaries. ROUGE-1 assesses basic vocabulary similarity between produced and reference summaries, based on unigram overlap. ROUGE-2 compares the bigram overlap, which accounts for phrase-level similarity and fluency. ROUGE-L compares the structure of the sentences in two texts, showing the longest common subsequence. The metrics are each reported as an F1 score (harmonic mean of precision and recall), which is a balanced measure of the amount of the reference summary captured and the amount of the generated summary relevant. ROUGE was chosen as the main evaluation measure as it is widely used as the standard evaluation measure for summarization research (Lin, 2004) and can be compared directly with other published results on news datasets from other summarization systems evaluated on news datasets.

5. Experiment Result

This section describes the full implementation of the NewsBrief system across two components — the Kaggle notebook, which covers the NLP pipeline from data loading through to ROUGE evaluation, and the Streamlit web application, which demonstrates the system's real-world deployment. Together, these components demonstrate how the proposed solution was implemented, evaluated and made publicly accessible.

5.1 Kaggle Notebook

The core NLP pipeline was developed and executed on Kaggle using a Tesla T4 GPU, leveraging the free cloud computing environment to run the facebook/bart-large-cnn model efficiently.

Kaggle notebook: https://github.com/KBVarshini/News-Summariser/blob/main/kaggle-notebook/NLP_Project.ipynb

5.1.1 Dataset

The Inshorts News Summary dataset was obtained from Kaggle (Sunnysai12345, 2017), covering Indian and international news sourced from The Hindu, Times of India and The Guardian, published between February and August 2017. It consists of two files:

1) **news_summary.csv**: 4,514 rows with columns: author, date, headlines, read_more, text and ctext, where ctext is the full article and text is the Inshorts reference summary. This file was retained as it contains genuine article-summary pairs.
2) **news_summary_more.csv**: 98,401 rows containing only headlines and short text snippets with no full article content, and was therefore excluded.

After removing rows with missing content, the final cleaned dataset comprised **4,396 article-summary pairs**.

author	date	headlines	read_more	text	ctext
Chhavi Tyagi	03 Aug 2017,	Daman & Diu	http://www.hindustantimes.com/indi	The Administration of Union Territory Dama	The Daman and Diu administration on Wednesday withdrew a circular that asked women staff to tie rakhis on male colleagues after the order triggered a backlash from employees ar
Daisy Mowke	03 Aug 2017,	Malaika slam	http://www.hindustantimes.com/boll	Malaika Arora slammed an Instagram user v	From her special numbers to TV?appearances, Bollywood actor Malaika Arora Khan has managed to carve her own identity. The actor, who made her debut in the Hindi film industry v
Arshiya Chop	03 Aug 2017,	'Virgin' now c	http://www.hindustantimes.com/patr	The Indira Gandhi Institute of Medical Scier	The Indira Gandhi Institute of Medical Sciences (IGIMS) in Patna amended its marital declaration form on Thursday, replacing the word ?virgin? with ?unmarried? after
Sumedha Sel	03 Aug 2017,	Aaj aapne pa	http://indiatoday.intoday.in/story/ab	Lashkar-e-Taiba's Kashmir commander Abu	Lashkar-e-Taiba's Kashmir commander Abu Dujana was killed in an encounter in a village in Pulwama district of Jammu and Kashmir earlier this week. Dujana, who had managed to gl
Aarushi Mahe	03 Aug 2017,	Hotel staff to	http://indiatoday.intoday.in/story/sex	Hotels in Maharashtra will train their staff t	Hotels in Mumbai and other Indian cities are to train their staff to spot signs of sex trafficking such as frequent requests for bed linen changes or a "Do not disturb" sign left on the doo
Sonu Kumari	03 Aug 2017,	Man found de	http://www.hindustantimes.com/delh	A 32-year-old man on Wednesday was foun	An alleged suspect in a kidnapping case was found hanging inside the washroom of the Jahangirpuri police station in north Delhi on Wednesday, hours after he was called by the cops
Parmeet Kau	03 Aug 2017,	Delhi HC red	http://indiatoday.intoday.in/story/del	The Delhi High Court reduced the compens	In an interesting ruling, the Delhi high court reduced the compensation awarded to a motor accident victim by 45 per cent after it found negligence on the part of both the parties.De
Chhavi Tyagi	03 Aug 2017,	60-yr-old lyn	http://www.hindustantimes.com/indi	A 60-year old Dalit woman was allegedly lyr	A 60-year old Dalit woman was allegedly lynched in Agra after villagers thought she was out to cut the hair of sleeping women, the first reported fatality of what appears to be turning
Parmeet Kau	03 Aug 2017,	Chopper flyir	http://www.hindustantimes.com/mun	An inquiry by the Aircraft Accident Investiga	Two years after a helicopter crash near the Bombay High offshore oil field killed two pilots, an inquiry by the Air Accident Investigation Bureau (AAIB) found that the chopper was flyin
Sumedha Sel	03 Aug 2017,	Congress ope	http://indiatoday.intoday.in/story/luc	The Congress party has opened a bank calle	It sounds like satire, but make no mistake: at a time when tomatoes have become forbiddingly pricey, the Congress Party has opened a 'State Bank of Tomato' in Uttar Pradesh's capit

*Figure 5.1.1.1: Sample Article-Summary Pairs from **news_summary.csv***

headlines	text
upGrad learner switches to career in ML & AI with 90% salary hike	Saurav Kant, an alumnus of upGrad and IIIT-B's PG Program in Machine learning and Artificial Intelligence, was a Sr Systems Engineer at Infosys with almost 5 years of work experience. The program and upGrad's 360-degree
Delhi techie wins free food from Swiggy for one year on CRED	Kunal Shah's credit card bill payment platform, CRED, gave users a chance to win free food from Swiggy for one year. Pranav Kaushik, a Delhi techie, bagged this reward after spending 2000 CRED coins. Users get one CRED
New Zealand end Rohit Sharma-led India's 12-match winning streak	New Zealand defeated India by 8 wickets in the fourth ODI at Hamilton on Thursday to win their first match of the five-match ODI series. India lost an international match under Rohit Sharma's captaincy after 12 consecutive
Aegon life iTerm insurance plan helps customers save tax	With Aegon Life iTerm Insurance plan, customers can enjoy tax benefits on your premiums paid and save up to √¢¬¬46,800^ on taxes. The plan provides life cover up to the age of 100 years. Also, customers have options to
Have known Hirani for yrs, what if MeToo claims are not true: Sonam	Speaking about the sexual harassment allegations against Rajkumar Hirani, Sonam Kapoor said, "I've known Hirani for many years...What if it's not true, the [#MeToo] movement will get derailed." "In the #MeToo movement,
Rahat Fateh Ali Khan denies getting notice for smuggling currency	Pakistani singer Rahat Fateh Ali Khan has denied receiving any notice from the Enforcement Directorate over allegedly smuggling foreign currency out of India. "It would have been better if the authorities would have served
India get all out for 92, their lowest ODI total in New Zealand	India recorded their lowest ODI total in New Zealand after getting all out for 92 runs in 30.5 overs in the fourth ODI at Hamilton on Thursday. Seven of India's batsmen were dismissed for single-digit scores, while their numbe
Govt directs Alok Verma to join work 1 day before his retirement	Weeks after ex-CBI Director Alok Verma told the Department of Personnel and Training to consider him retired, the Home Ministry asked him to join work on the last day of his fixed tenure as Director on Thursday. The minist
Called PM Modi 'sir' 10 times to satisfy his ego: Andhra CM	Andhra Pradesh CM N Chandrababu Naidu has said, "When I met then US President Bill Clinton, I addressed him as Mr Clinton, not as 'sir'. (PM Narendra) Modi is my junior in politics...I addressed him as sir 10 times." "I did t
Cong wins Ramgarh bypoll in Rajasthan, takes total to 100 seats	Congress candidate Shafia Zubair won the Ramgarh Assembly seat in Rajasthan, by defeating BJP's Sukhwant Singh with a margin of 12,228 votes in the bypoll. With this victory, Congress has taken its total to 100 seats in the
UP cousins fed human excreta for friendship with boys	Two minor cousins in Uttar Pradesh's Gorakhpur were allegedly repeatedly burnt with tongs and forced to eat human excreta by their family for being friends with two boys from the same school. The cousins revealed their or
81-yr-old woman conducts physical training in J'khand schools	Isha Ghosh, an 81-year-old member of Bharat Scouts and Guides (BSG), has been imparting physical and mental training to schoolchildren in Jharkhand for several decades. Chaibasa-based Ghosh reportedly walks seven k
Ram, Krishna didn't smoke, why should we: Ramdev to sadhus at Kumbh	Urging saints and seers at the Kumbh Mela to quit smoking, Yoga guru Ramdev said, "We follow Ram and Krishna who never smoked in their life then why should we?" Making them take a pledge to quit tobacco, he collected c
Pharma exec gave doctor a lap dance to sell medicine in US: Witness	Former stripper and regional sales director of a pharmaceutical company, Sunrise Lee, gave a doctor a lap dance in a nightclub to persuade him to prescribe an addictive fentanyl spray in 2012, the company's sales represer
I only cried at my 'bidaai' as I felt peer pressure: Isha Ambani	Reliance Industries' Chairman Mukesh Ambani's daughter Isha Ambani, who got married last month, said she only cried at her 'bidaai' because she felt peer pressure as everyone was crying, especially her parents. "I was em
Louis Vuitton owner to stockpile 4 months of wine, spirits in UK	Louis Vuitton owner LVMH, which makes high-end beverages like Mo√É¬´t & Chandon champagne and Hennessy cognac, said it's stockpiling four months' worth of wine and spirits in UK in preparation for Brexit. "We're ready
Karan Johar, Tabu turn showstoppers on opening night of LFW	Filmmaker Karan Johar and actress Tabu turned showstoppers for Gaurav Gupta on the opening night of Lakm√É¬© Fashion Week Summer/ Resort 2019. While Johar wore a red sequinned jacket with black pants, Tabu walke
Those on bail will go to jail: PM Modi takes jibe at Rahul	In a jibe at Congress President Rahul Gandhi, PM Narendra Modi on Wednesday said those on "bail will have to go to jail." PM Modi added, "He is out on bail and his associates too are facing charges. I know they will be convi

*Figure 5.1.1.2: Sample Article-Summary Pairs from **news_summary_more.csv***

5.1.2 Preprocessing

Raw news articles required cleaning before being fed into the BART tokenizer. A clean_text() function was implemented to perform the following seven preprocessing steps:

1. **String conversion** — Input converted to string to handle null values
2. **HTML tag removal** — Tags matching <[^>]+> are removed
3. **HTML entity removal** — Entities such as & and   are stripped
4. **URL removal** — Links matching https?://\S+ and www\.\S+ are removed
5. **Unicode normalisation** — Text normalised using NFKD and re-encoded to ASCII
6. **Non-ASCII removal** — Remaining non-ASCII bytes replaced with whitespace
7. **Whitespace collapsing** — Multiple spaces, tabs and newlines collapsed into a single space

The same function was applied to both articles and reference summaries to ensure consistency across inputs and evaluation targets.

5.1.3 Exploratory Data Analysis

Prior to model inference, exploratory data analysis was conducted to understand the dataset characteristics. As shown in Figure 5.1.3.1, article lengths vary significantly with a mean of 343 words and a maximum of 12,202 words, while reference summaries are tightly clustered between 44 and 62 words with a mean of 58 words, reflecting Inshorts' strict editorial constraint. The compression ratio of approximately 0.17 confirms that summaries capture only 17% of the original article content on average, highlighting the significant content reduction required from the summarization model.

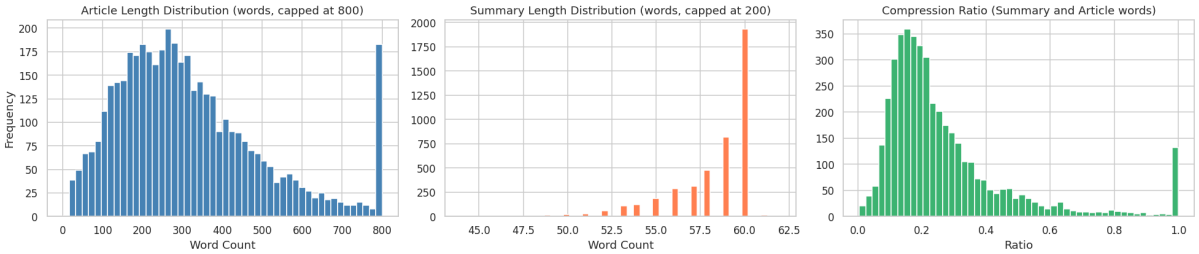


Figure 5.1.3.1: Article Length Distribution, Summary Length Distribution and Compression Ratio, Inshorts News Summary Dataset (n=4,396)

```
-------- Article Length Stats (Words) -------
count     4396.0
mean       343.1
std        357.6
min          1.0
25%        185.0
50%        283.0
75%        411.0
max      12202.0

------- Summary Length Stats (Words) -------
count    4396.0
mean       58.3
std         2.3
min        44.0
25%        57.0
50%        59.0
75%        60.0
max        62.0
```

Figure 5.1.3.2: Article and Summary Length Statistics

5.1.4 Tokenisation

Before model inference, the BART tokenizer was loaded from HuggingFace Hub using BartTokenizer.from_pretrained('facebook/bart-large-cnn') with a vocabulary size of 50,265 tokens. Token length analysis was conducted on a working sample of 1,000 articles to verify compatibility with BART's maximum input limit of 1,024 tokens. As shown in Figure 5.1.4.1, the majority of articles fall well within this limit, with only 0% of sampled articles requiring truncation — confirming that the dataset is well-suited for direct inference without significant information loss.

Key tokenizer special tokens:

- **BOS token** <s> - marks the beginning of sequence.
- **EOS token** </s> - marks the end of sequence.
- **PAD token** <pad> - pads shorter sequences for uniform batch processing.

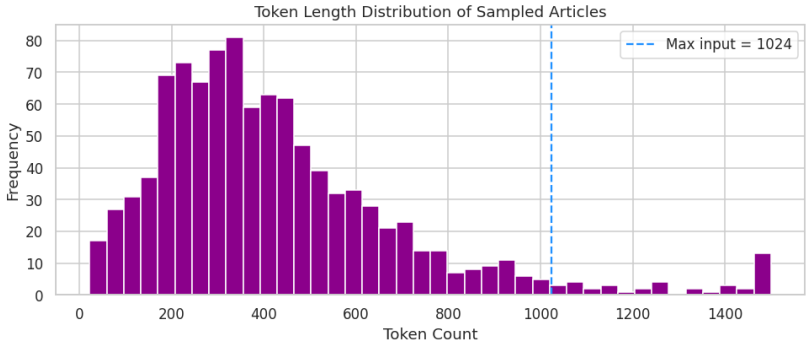


Figure 5.1.4.1: Token Length Distribution of Sampled Articles

5.1.5 Model Inference

The facebook/bart-large-cnn model was loaded using BartForConditionalGeneration.from_pretrained() and moved to the Tesla T4 GPU. With 406 million parameters, the model was run in evaluation mode to disable dropout layers during inference. A summarize_article() function tokenizes each article, generates a summary using beam search and decodes the output back to readable text. The key hyperparameters used are summarised below:

Parameter	Value	Purpose
max_input_tokens	1024	Maximum tokens fed to encoder
min_summary_len	30	Prevents excessively short summaries
max_summary_len	150	Caps summary length

beam_size	4	Explores 4 hypotheses simultaneously
length_penalty	2.0	Favours longer summaries
no_repeat_ngram	3	Prevents 3-gram repetition

Table: 5.1.5.1: Parameter settings of the model.

Inference was conducted on 1,000 randomly sampled articles (random_state=42), completing in approximately 20 minutes at 1.22 seconds per article.

5.1.6 ROUGE Evaluation

ROUGE evaluation was conducted by comparing each generated summary against its corresponding Inshorts reference summary using the rouge_scorer library with stemming enabled. Table 1 presents the average ROUGE F1 scores across all 1,000 evaluated articles.

Metric	F1 Score	Percentage	Best Score	Worst Score
ROUGE-1	0.4244	42.44%	0.8855	0.0440
ROUGE-2	0.2056	20.56%	0.7600	0.0000
ROUGE-L	0.3052	30.52%	0.8454	0.0345

Table 5.1.6.1: ROUGE Evaluation Results (n=1,000)

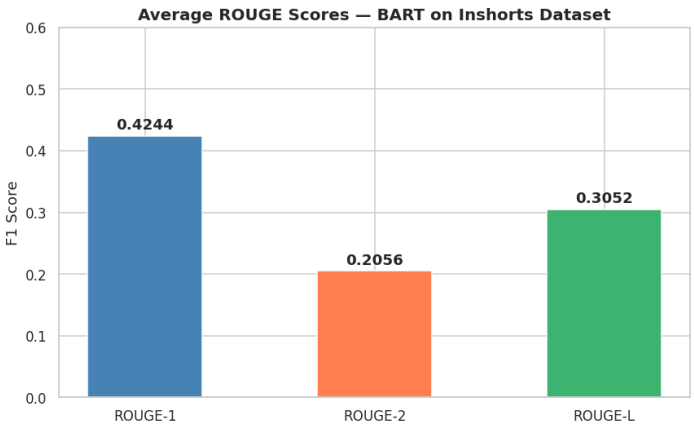


Figure 5.1.6.1: Average ROUGE Scores Bar Chart

In Figure 5.1.6.1, the ROUGE-1 score of 42.44% indicates strong word-level overlap between generated and reference summaries. The ROUGE-2 score of 20.56% reflects moderate phrase-level similarity, expected for abstractive summarization since the model rephrases rather than copies. The ROUGE-L score of 30.52% confirms reasonable sentence-level structural alignment. A strong correlation of 0.881 between ROUGE-1 and ROUGE-L scores confirms both metrics consistently agree on summary quality across the dataset.

Best and Worst Performing Summaries

Table 5.1.6.2 presents sample generated summaries against their Inshorts reference summaries, illustrating both the strongest and weakest performing cases.

Headline	Reference Summary (Inshorts)	Generated Summary (BART)	ROUGE-1
China building high-speed rail tunnel below Great Wall	China is reportedly constructing a 12-km-long high-speed train tunnel through one of the portions of its iconic Great Wall to connect Beijing with Zhangjiakou city which will co-host 2022 Winter Olympics. The line is designed for trains traveling up to 350 kmph and is expected to shorten the single-journey time from more than 3 hours to approximately one hour.	China is constructing a 12-km-long high-speed train tunnel through one of the portions of its iconic Great Wall. The tunnel will connect Zhangjiakou city which is due to co-host 2022 winter Olympics along with Beijing. The line is designed for trains travelling up to 350km per hour, which will shorten the single-journey time from more than 3 hours to approximately one hour.	0.8855
Christopher Nolan's Dunkirk hits theatres	Director Christopher Nolan's film Dunkirk which released on Friday, "is one of the greatest war movies ever made," wrote Hindustan Times. The Guardian wrote, "It's a visceral piece of film-making," while Times of India called it "the first real contender for Best Picture" at Oscars. The film has been rated 5/5 (Hindustan Times, The Guardian) and 4.5/5 (Times Of India).	Dunkirk is an existential masterpiece set across three parallel plots destined to collide. Tom Hardy, Fionn Whitehead, Mark Rylance, Harry Styles, Cillian Murphy, Kenneth Branagh.	0.0440
What becomes more expensive after Union Budget 2017?	Finance Minister Arun Jaitley on Wednesday said excise duty on unmanufactured tobacco will be doubled to 8.3% under Union Budget 2017. Other tobacco-related products like cigars, cheroots, pan masala, and cigarettes will also cost more. Prices of items like roasted or salted cashew nuts, circuit boards used in smartphones, and RO membrane elements for household water filters will also increase.	Personal income tax in Rs 2.5 lakh - 5 lakh bracket has been reduced to 5 per cent from 10 per cent. Political parties receiving donations above Rs 2,000 will now have to disclose the identity of the donor.	0.0784

Table 5.1.6.2: Sample Generated Summaries vs Reference Summaries

The best performing summaries demonstrate near-identical meaning to reference summaries, confirming BART's strong performance on straightforward factual articles. The worst performing cases reflect the multi-topic challenge identified in the Background section — where BART selected a different aspect of the article than the human editor, resulting in low lexical overlap despite producing a factually correct summary.

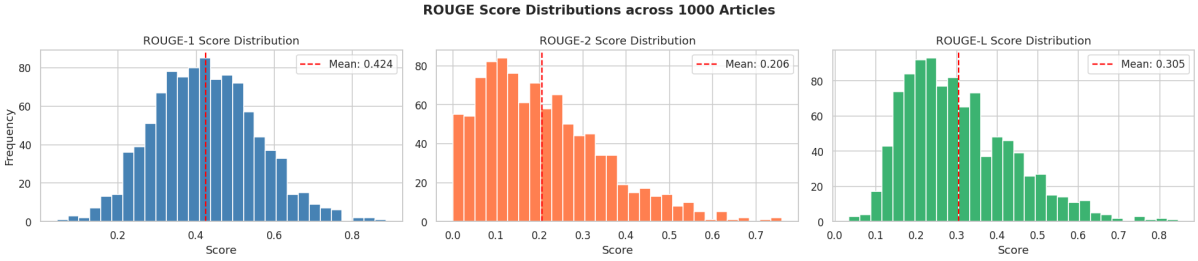


Figure 5.1.6.2: ROUGE Score Distributions

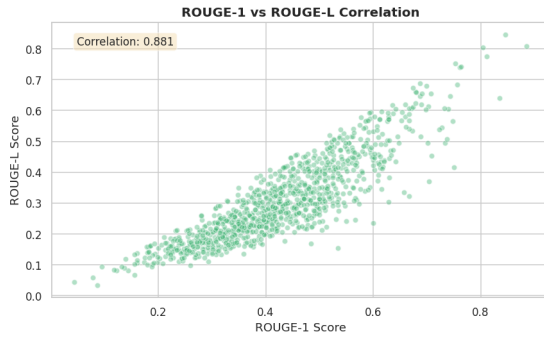


Figure 5.1.6.3: ROUGE-1 VS ROUGE-L Correlation

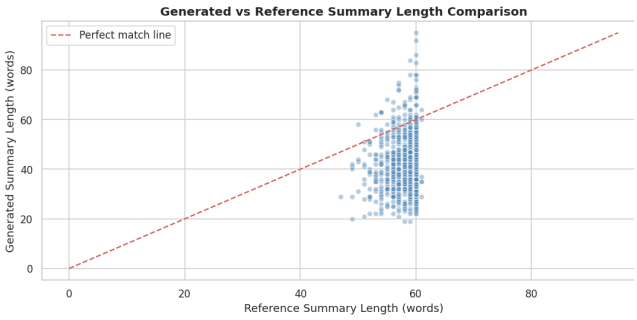


Figure 5.1.6.4: Generated vs Reference Summary Length Comparison

5.2 Web Application

The NewsBrief web application was developed using Streamlit and deployed publicly via Streamlit Cloud.

GitHub : https://github.com/KBVarshini/News-Summariser

5.2.1 System Pipeline

Figure 5.2.1.1 below illustrates the end-to-end workflow of the NewsBrief application, supporting two input modes — fetching live news via NewsAPI and summarizing custom user-pasted text — both routed through the HuggingFace BART inference pipeline before rendering as Inshorts-style news cards.

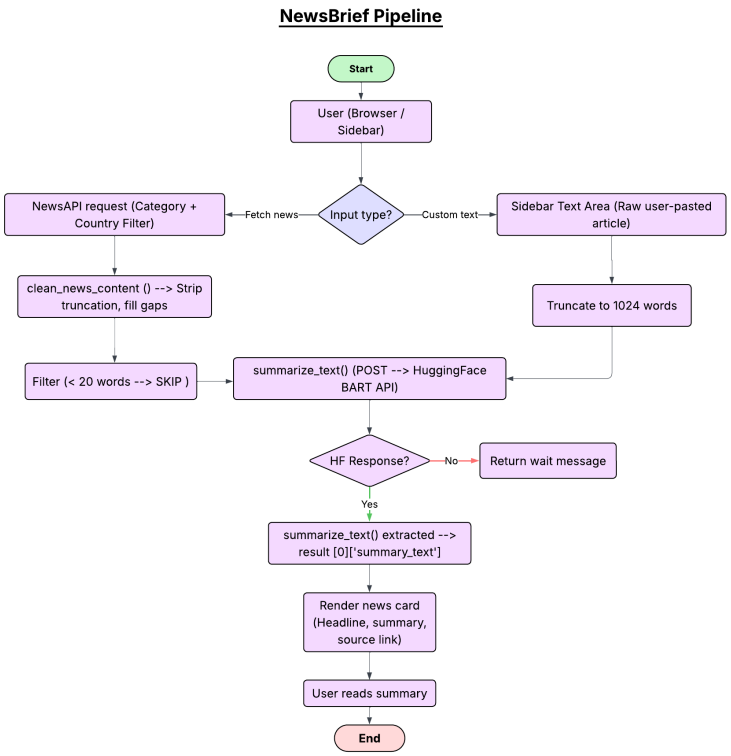


Figure 5.2.1.1: NewsBrief System Pipeline, End-to-end workflow from user input to summary display

Based on Figure 5.2.1.1, the NewsBrief app begins when the user opens it in their browser. From there, the user can interact in one of two ways, either fetching live news through the main page or pasting their own text in the sidebar.

If the user chooses to fetch news, they select a category and country from the sidebar dropdowns and click "Fetch Today's News." The app sends a request to NewsAPI using those filters and retrieves a list of articles. Each article is then cleaned to remove unwanted characters and fill any content gaps. Any article that is shorter than 20 words is skipped and not displayed.

If the user instead pastes their own article into the sidebar text box and clicks Summarize, the app takes that raw text and immediately trims it down to a maximum of 1024 words before doing anything else.

Both paths then lead to the same point - the processed text is sent to the HuggingFace BART API for summarization. If the API does not respond, the app returns a wait message to the user rather than crashing. If the API responds successfully, the summary is extracted from the response.

Finally, the result is displayed as a news card containing the headline, the AI-generated summary and a link to the full article. The user reads the summary and the process is complete.

5.2.2 Application Features

The application provides the following key features:

- **Live News Fetching**: Fetches today's top headlines via NewsAPI across multiple categories (general, technology, business, science, health, sports, entertainment) and countries, with each article automatically summarized using BART.

- **Custom Text Summarization**: Users can paste any article text directly into the sidebar for instant summarization.

- **News Cards:** Each summarized article is displayed as an Inshorts-style card showing the headline, AI-generated summary and source link.

- **Country Selector**: Users can switch between multiple countries including the United States, United Kingdom, Australia and India.

Figure 5.2.2.1 and Figure 5.2.2.2 are the initial version of the NewsBrief app features a simple, functional layout. The overall design is minimal with little visual styling, serving as the working foundation before any UI enhancements were applied.

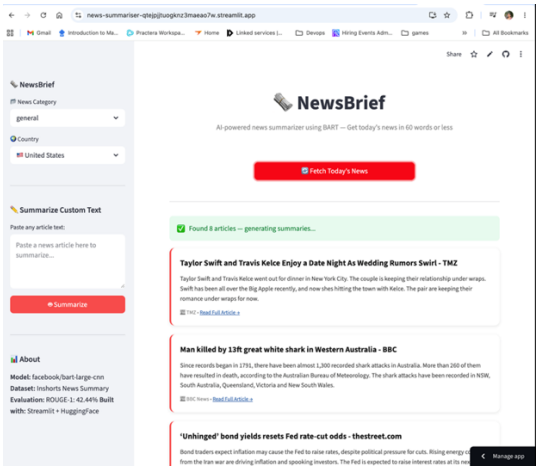


Figure 5.2.2.1: Mockup NewsBrief Home Screen

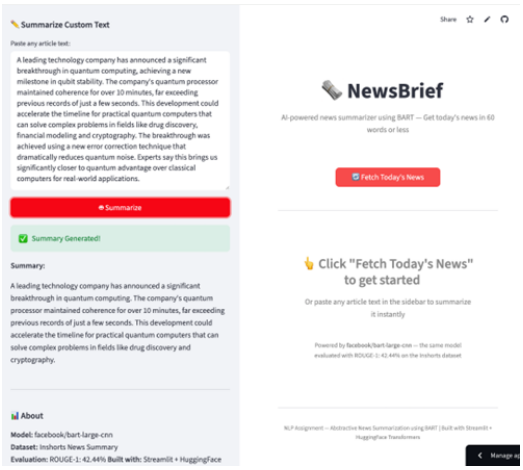


Figure 5.2.2.2: Mockup Custom Text Summarizer

Figure 5.2.2.3 and Figure 5.2.2.4 are the enhanced version of the NewsBrief app reflects a significantly more polished and cohesive visual design. The sidebar adopts a purple colour scheme with a branded "NewsBrief AI-Powered Summaries" header, cleaner dropdown styling and a structured model info panel at the bottom. The main page introduces a newspaper-style masthead featuring the NewsBrief logo in a serif font, the current date, and edition details aligned to the right. The "Fetch Today's News" button is restyled in a deep purple to match the overall theme. The layout feels more professional and intentional, closely resembling a editorial news interface compared to the plain functional look of the initial mockup.

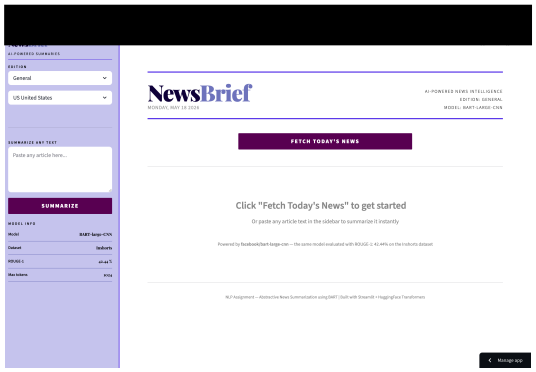


Figure 5.2.2.3: Final NewsBrief Home Screen



Figure 5.2.2.4: Final Custom Text Summarizer

5.2.3 System Testing

The NewsBrief web application was tested against five defined test cases to validate system behaviour across different usage scenarios, as shown in Table 2.

Test Case ID	Testing Details	What to check?
T1	Open the app, don't click anything.	Empty state shows (“Today's edition awaits”), no crash.
T2	Click "Fetch today's news" with default settings (General, Australia).	Cards appear, lead story is full-width, rest in 2 col grid.
T3	Paste a paragraph of text in the sidebar → click Summarize.	Summary appears in the red-bordered box below, no error.
T4	Paste only 3 words → click Summarize.	Still returns something (short text goes straight to BART, no filter on sidebar path).
T5	Switch to a non-Australian country (e.g. India) → Fetch.	Articles are India-relevant, app doesn't crash on the everything endpoint fallback.
T6	No internet connection	Graceful error message shown
T7	Paste article > 1024 words	Truncated correctly, summary generated

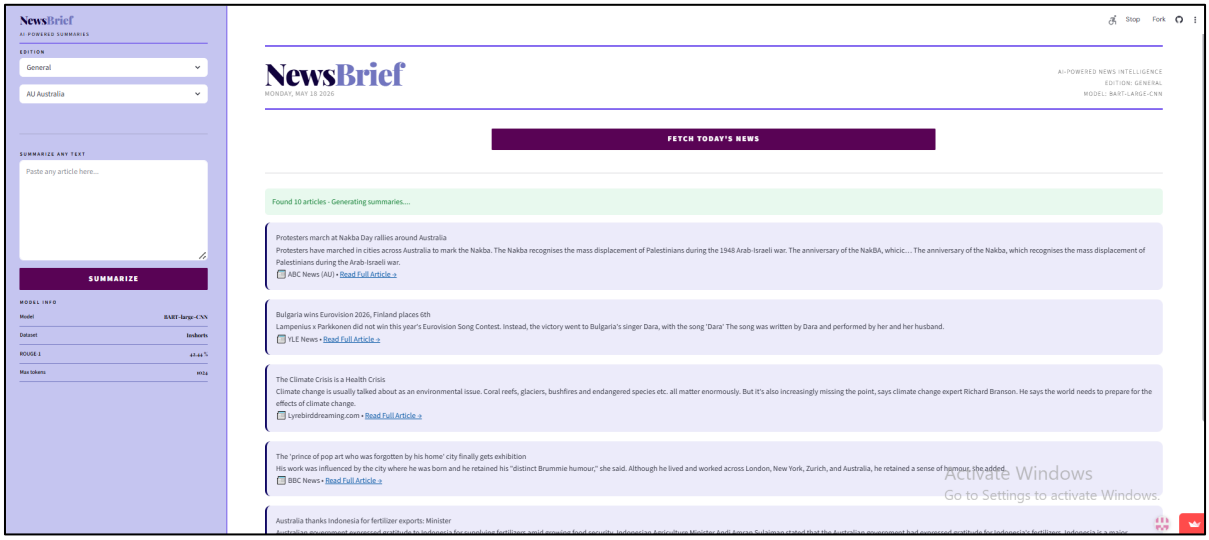
Table 5.2.3.1: List of Manual Test Cases.

1) Test Case 1 (T1):



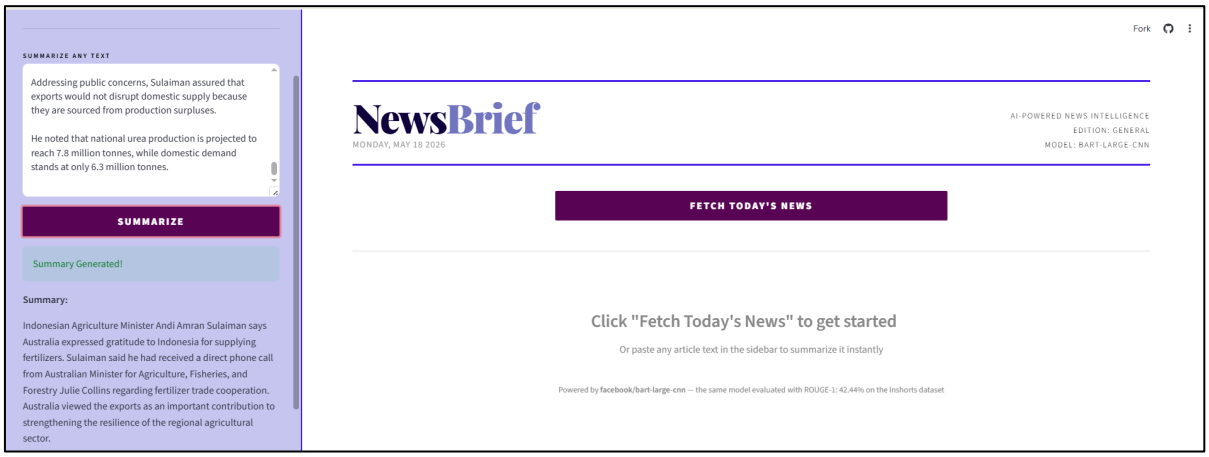
This test case shows that when the application is first launched, the user is presented with the NewsBrief interface. No actions are taken. The expected outcome is that the app loads successfully, displaying the empty state message prompting the user to fetch news, confirming the app launches without errors.

2) Test Case 2 (T2):



The user opens the app with the default settings, category set to General and country set to Australia. The user clicks the "Fetch Today's News" button. The app retrieves the latest Australia news articles and displays them as cards, with the lead story appearing in full width and the remaining articles arranged in a two-column grid. The screenshot above confirms that the *fetch_news()* function is able to retrieve news based on the selected country only.

3) Test Case 3 (T3):



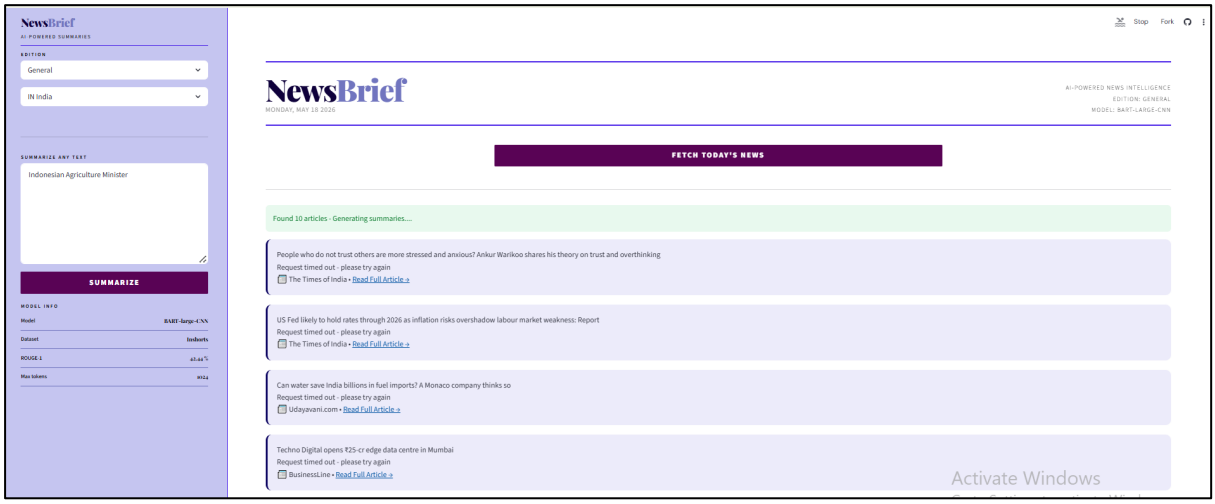
The user copies a full paragraph of text and pastes it into the "Summarize any text" box in the sidebar. After clicking the Summarize button, the app processes the text and displays the generated summary in the result box below. It also displays a message, “Summary Generated!”. The expected outcome is that the summary appears without any errors. From the screenshot above, we can see that the *summarize_text()* function is working successfully.

4) Test Case 4 (T4):



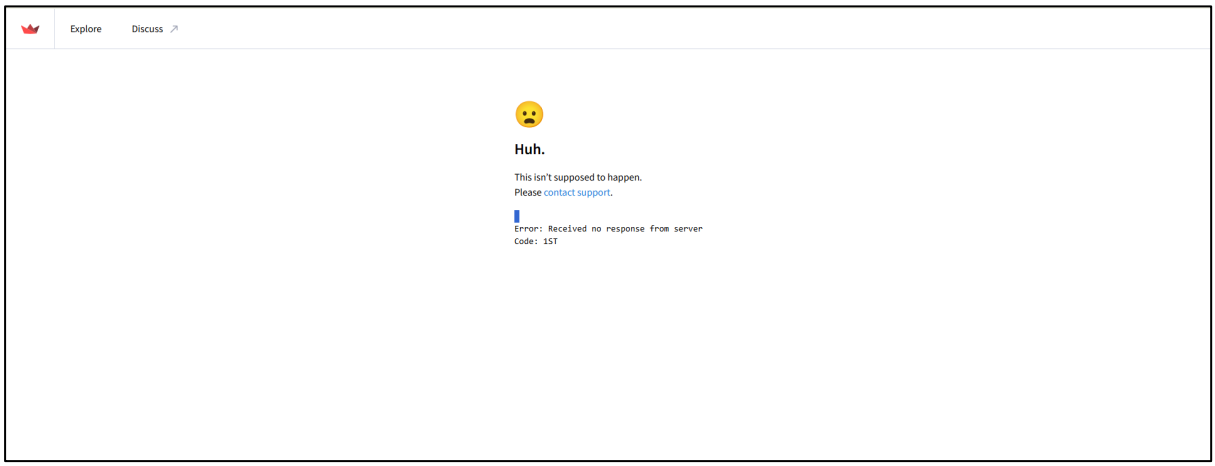
The user pastes a short piece of text containing fewer than 50 words into the sidebar text box and clicks Summarize button. Rather than processing the input, the app displays a validation message - "*Please enter at least 50 words. You entered 7 word(s).*", informing the user that the input is too short to summarize.

5) Test Case 5 (T5):



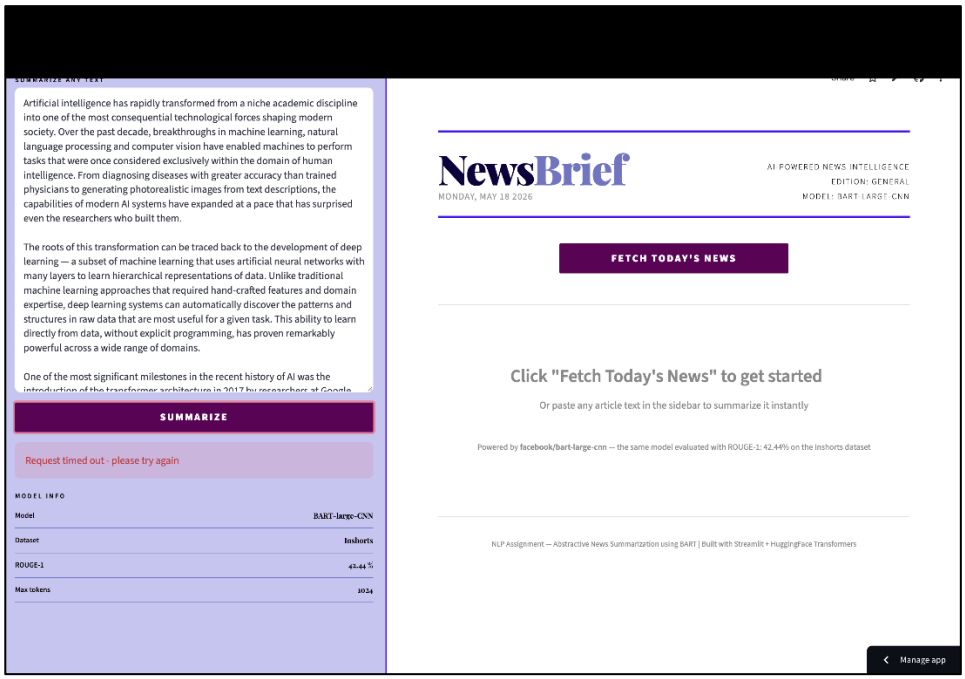
The user changes the country setting in the sidebar to a non-Australian country, such as India, and clicks Fetch Today's News. The app retrieves and displays articles relevant to the selected country, confirming that the country filter works correctly and the app remains stable throughout.

6) Test Case 6 (T6):



With no active internet connection, the user attempts to access the app. It fails to load entirely. The Streamlit Cloud hosting page displays an unexpected error - *"Error: Received no response from server"*, indicating that without connectivity, the app cannot be reached at all and no custom error handling is triggered.

7) Test Case 7 (T7):



The user pastes a lengthy article exceeding 1024 words into the sidebar text box and clicks Summarize button. The app attempts to process and truncate the input before sending it to the HuggingFace BART API. However, due to the size of the request, the API takes too long to respond and the app displays a timeout message - *"Request timed out - please try again"*, indicating that while the truncation works correctly, the summarization request exceeded the API's response time limit.

Conclusion

This project set out to answer a fundamental question raised in the Introduction: *how can people stay updated with the world without having to read news articles word by word?* Through the development of NewsBrief, an abstractive news summarization system based on the BART transformer model, this project demonstrates that modern NLP provides a compelling and practical answer to that question.

Unlike extractive approaches, the system generates concise, human-readable summaries while preserving the meaning and context of the original article. Evaluated using ROUGE metrics on the Inshorts News Summary dataset, the system achieved a ROUGE-1 of 42.44%, ROUGE-2 of 20.56% and ROUGE-L of 30.52% across 1,000 articles — demonstrating effective summarization performance on real-world news content. The deployment of NewsBrief as a publicly accessible Streamlit web application further highlights its practical applicability for real-time news summarization.

Although the system performed well, limitations such as domain mismatch and occasional factual inconsistencies were observed. Future work could focus on domain-specific fine-tuning, improved evaluation methods and lightweight model optimisation. Overall, this project demonstrates the effectiveness of transformer-based abstractive summarization for improving news accessibility and reducing information overload.

Team Member Contribution

Team member 1:

1. Developed the Kaggle notebook covering BART model loading, inference and ROUGE evaluation.
2. Conducted the literature review and background research on abstractive news summarization.
3. Built the Streamlit web application with NewsAPI integration and sidebar features.
4. Set up the GitHub repository and deployed the application on Streamlit Cloud.
5. Collaborated on troubleshooting bugs, resolving issues during development and deployment.
6. Contributed to report writing and oral presentation preparation.

Brunda Varshini Katapalli :

1. Developed the Kaggle notebook covering BART model configuration, pre-processing, Exploratory Data Analysis (EDA) and Tokenisation.
2. Conducted the literature review and background research on abstractive news summarization.
3. Built the model's pipeline.
4. Enhanced the GUI layout, styling, and overall visual presentation of the app, NewsBrief.
5. Contributed to troubleshooting issues during development and deployment.
6. Contributed to report writing and oral presentation preparation.

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